Swastik Chowbay

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PERSONAL

Nationality Indian
Languages English, Hindi, Bengali, Assamese; Spanish/Italian (elementary)

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow July 2024 – Present
Università degli Studi di Milano, Milan, Italy

Postdoctoral Researcher March 2024 – June 2024
Indian Institute of Astrophysics, Bengaluru, India

Intern January 2021 – April 2021
European Southern Observatory, Santiago, Chile

EDUCATION

Ph.D. in Astrophysics July 2018 – March 2024
Indian Institute of Astrophysics, Bengaluru

M.Sc. in Physics July 2016 – May 2018
University of Hyderabad, Hyderabad *(CGPA: 8.67/10)*

B.Sc. in Physics July 2013 – May 2016
University of Calcutta, Kolkata *(70.12%)*

SKILLS

Languages Python (primary), IDL, R, Java, C, C++, Fortran
Software iSpec, IRAF, MOOG, Emcee, RADMC-3D, IRDAP, CASA (elementary)
Systems Linux, macOS, Windows

ACHIEVEMENTS

- Best poster award, *Pressing for Progress: Gender Equity in Physics* 2019
- Qualified NET-JRF (PhD eligibility, India) 2017
- Top 15 of 11,535 in National Graduate Physics Examination 2016

PUBLICATIONS

Summary: 11 refereed publications | 6 as first author | 6 manuscripts in preparation
Google Scholar: **205** citations | *h*-index: **7** | i10-index: **7** *(as of May 2026)*

Major research contributions

- **Stellar hosts of directly imaged planets and brown dwarfs:** showed that wide-orbit giant planets do not require metal-rich hosts, indicating a formation pathway distinct from close-in giants.
- **Galactic chemical evolution as a driver of planet demographics:** demonstrated that exoplanet host-star abundance and age trends naturally arise from Galactic chemical evolution, with high-mass-planet hosts being systematically younger.
- **Star-hopping high-contrast imaging:** delivered the first artefact-free polarimetric and total-intensity SPHERE images of the LkCa 15 disk, achieving a $\sim$2-mag contrast improvement over classical ADI/RDI.

First-author and lead-contributor publications

1. *Imaging the LkCa 15 system in polarimetry and total intensity without self-subtraction artefacts.* **C. Swastik**, Z. Wahhaj, M. Benisty, S. Arora, C. Ginski, B. B. Ren, R. G. van Holstein, R. de Rosa, R. K. Banyal, R. Tazaki. *Astronomy & Astrophysics* (2026, in press).

2. *Age analysis of exoplanet hosting stars from isochrone models.* **C. Swastik**, R. K. Banyal, M. Narang, A. Unni, T. Sivarani. *Astronomical Journal* **167**, 270 (2024).

3. *Age distribution of exoplanet host-stars: chemical and kinematic age proxies from Gaia DR3.* **C. Swastik**, R. K. Banyal, M. Narang, A. Unni, B. Banerjee, P. Manoj, T. Sivarani. *Astronomical Journal* **166**, 91 (2023).

4. *Galactic chemical evolution of exoplanet host-stars: are high-mass planetary systems young?* **C. Swastik**, R. K. Banyal, M. Narang, P. Manoj, T. Sivarani, S. P. Rajaguru, A. Unni, B. Banerjee. *Astronomical Journal* **164**, 60 (2022).

5. *Host-star metallicity of directly imaged wide-orbit planets: implications for planet formation.* **C. Swastik**, R. K. Banyal, M. Narang, P. Manoj, T. Sivarani, B. E. Reddy, S. P. Rajaguru. *Astronomical Journal* **161**, 114 (2021).

6. *Bouncing Universe with exotic radiation.* **C. Swastik**, P. K. Suresh, B. Maity. *Reports in Advances of Physical Sciences* **4**, 2050001 (2020). *(M.Sc. thesis project; theoretical cosmology.)*

Co-authored publications

7. *The 10 pc neighborhood of habitable zone exoplanetary systems: threat assessment from stellar encounters & supernovae.* T. Pyne, R. K. Banyal, **C. Swastik**, A. De. *Astronomical Journal* **169**, 13 (2025).

8. *Exoplanet classification through Vision Transformers with temporal image analysis.* A. Choudhary, S. Bandari, B. S. Kushvah, **C. Swastik**. *Astronomical Journal* (2025, accepted).

9. *PDS 70 unveiled by star-hopping: total intensity, polarimetry, and millimeter imaging modeled in concert.* Z. Wahhaj, M. Benisty, C. Ginski, S. Arora, **C. Swastik**, R. G. van Holstein, R. de Rosa, B. Yang, J. Bae, B. Ren. *Astronomy & Astrophysics* **687**, A257 (2024).

10. *Protoplanetary disks in K_s-band total intensity and polarized light.* B. B. Ren, M. Benisty, C. Ginski, R. Tazaki, N. L. Wallack, J. Milli, A. Garufi, J. Bae, S. Facchini, F. Ménard, P. Pinilla, **C. Swastik**, R. Teague, Z. Wahhaj. *Astronomy & Astrophysics* (2024).

11. *Carbon abundance of stars in the LAMOST–Kepler field.* A. Unni, M. Narang, T. Sivarani, M. Puravankara, R. K. Banyal, A. Surya, S. P. Rajaguru, **C. Swastik**. *Astronomical Journal* **164**, 181 (2022).

MANUSCRIPTS IN PREPARATION

As lead author

1. *Host-star metallicities and kinematics of directly imaged brown-dwarf companions.* **C. Swastik**, R. K. Banyal, M. Muduli, S. Soni, A. Jino, S. Facchini, G. Lodato, Z. Wahhaj, T. Sivarani, G. Maheswar, P. Saraf, A. Choudhary, Bhavya A. K., N. Puthiyaveettil, B. Banerjee, A. Surya, S. Biswas.

2. *First optical spectroscopy of WISPIT 2: evidence for suppressed stellar accretion.* **C. Swastik**, T. Sivarani, R. K. Banyal, S. Facchini, Z. Wahhaj, Navaneet, M. Puravankara.

3. *ERIS/NIX L'-band vortex coronagraph observations of the transitional disks J1842 and V4046 Sgr.* **C. Swastik**, S. Facchini, D. Fedele, V. Roccatagliata, G. Cugno, V. Christiaens, Exo-ALMA team.

4. *A controlled comparison of confirmed exoplanet host samples from Kepler, K2, and TESS.* **C. Swastik**, S. Soni.

Co-authored

5. *Thorium abundances in FGK planet-host stars: Galactic chemical evolution and a planet-host signature.* A. Choudhary, P. Saraf, **C. Swastik**, B. S. Kushvah, R. K. Banyal, T. Sivarani, M. Gopinath.

6. *Star-hopping scattered-light imaging and combined ALMA–SPHERE modeling of the HD 163296 protoplanetary disk.* Navaneeth M. P., Z. Wahhaj, **C. Swastik**, R. K. Banyal.

SELECTED CONFERENCES AND MEETINGS

- COSPAR-2024, Busan, South Korea — *Oral* — July 2024
- Exoplanet Conference, IISER Pune — *Oral* — August 2023
- Towards Other Earths III, Porto, Portugal — *Poster* — July 2023
- Protostars and Planets VII, Kyoto, Japan — *Poster* — April 2023
- Planet–ESLAB 2023, ESA, Netherlands — *Poster* — March 2023
- 41st ASI Meeting, IIT Indore — *Oral* — March 2023

FUNDING & AWARDED RESOURCES

- **HPC compute time — Leonardo cluster (CINECA), 10,000 CPU-hours** — 2026
  PI; long-term monitoring and modelling of the WISPIT-2 host star.
- **JWST Cycle 5 GO — awarded as Co-I** (Program 11087) — 2026
  The first observation of H_2 absorption in a protoplanetary disk through transmission spectroscopy (NIRSpec IFU; PI: F. M. Alarcon).
- **VLT/SPHERE telescope time — ESO Cycle P116, awarded as PI** — 2025–2026
  - *Revealing The Moth's Sculptors: SPHERE's Star-Hopping Hunt for Planets and Disk Dynamics in HD 61005* (116.29D7).
  - *Unveiling Disk Substructures and Planet Formation: Scattered-Light Imaging of DoAr 33, DoAr 28, and AA Tau* (116.299M).
- **VLT/SPHERE telescope time — awarded as Co-PI** — 2024–2026
  - *Capturing dusty winds in the Orion proplyds* (116.292Y; PI: S. Facchini).
  - *Rings, spirals or planets? A first clear view of the HD 141569A system* (113.269Y; PI: Z. Wahhaj).
- **Himalayan Chandra Telescope (HCT) — 4 awarded PI programs over multiple cycles** — 2020–2026
  - *Anchoring an Accreting Protoplanet: the First Optical–NIR Spectroscopic Baseline of WISPIT-2* (HCT-2026-C2-P47).
  - *Long-term monitoring of the WISPIT-2 host star* (HCT 2026).
  - *Determination of host-star metallicity of directly imaged planets* (HCT-2020-C2-P34).
  - *Determination of host-star metallicity of directly imaged brown dwarfs* (HCT-2020-C3-P10).
- **DST International Travel Grant — INR 150,000** — 2023
  For attending *Towards Other Earths III*, Porto, Portugal.

OBSERVATIONAL EXPERIENCE

- Awarded ∼100 hours over three cycles on HFOSC/Himalayan Chandra Telescope, Hanle.
- HIP 41378f transit campaign: PI for India sites (GROWTH-India, Hanle; JCBT, Kavalur).
- Hands-on data reduction with VLT/SPHERE (IRDAP, IRDIS), spectroscopic pipelines (IRAF/PyRAF, Astropy), and high-contrast imaging post-processing.

STUDENTS MENTORED

- **Catherine John** (M.Sc., Christ University, Bengaluru) — *Demographics of exoplanets: analysis of the mass–radius relation.*
- **Aarthi Krishna** (M.Sc., IISER Tirupati) — *Understanding the brown-dwarf desert by analysing the host stars of directly imaged brown dwarfs.*
- **Athul Rathnakar** (M.Sc., St. Xavier's University, Bangalore) — *Investigating the stellar ages of FGK stars using stellar isochrone models.*
- **Satyam Soni** (M.Sc., NIT Rourkela) — *Investigating the brown-dwarf desert for directly imaged exoplanets.*

- **Tisyagupta Pyne** (M.Sc., Visva-Bharati University, Santiniketan) — *The 10 pc stellar neighbourhood of habitable zone planetary systems.*
- **Mritunjaya Muduli** (Visvesvaraya Technological University, Belagavi) — *TBD.*
- **Anupma Choudhary** (Indian Institute of Technology (Indian School of Mines), Dhanbad) — *TBD.*

OUTREACH & SERVICE

- Student representative on IIA's outreach committee; led sky-watch programs for underprivileged children and teacher-training programs for government school teachers around Bangalore (IAU-supported).
- Co-author of thematic proposal *New Worlds: Are we alone?*, selected under Open Astronomy Schools, IAU-100; volunteer with animal-welfare organizations.
- **Media coverage:** Research featured by India's Department of Science & Technology and the Ministry of Information & Broadcasting.

REFEREES

- **Dr. Stefano Facchini** — Postdoc Supervisor; Assoc. Prof., UniMi; *stefano.facchini@unimi.it*
- **Dr. Zahed Wahhaj** — Lead Collaborator; Staff Astronomer, ESO; *zwahhaj@eso.org*
- **Dr. Ravinder K. Banyal** — PhD Supervisor; Assoc. Prof., IIA, Bengaluru; *banyal@iiap.res.in*
- **Dr. Sivarani Thirupathi** — Collaborator; Professor, IIA, Bengaluru; *sivarani@iiap.res.in*
- **Dr. Maheswar Gopinath** — Doctoral Committee; Assoc. Prof., IIA, Bengaluru; *maheswar.g@iiap.res.in*